

View results

Respondent

30

Anonymous

37:42

Time to complete

A. General information about the respondent

1. * Country

Sweden



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Government offices Sweden, Ministry for Education and Research

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

<https://www.regeringen.se/sveriges-regering/utbildningsdepartementet/>

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

7. * Full name of officially designated contact person(s) who completed this survey

8. * Position of officially designated contact person(s) who completed this survey

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

11. Brief description of the consultation process established for the preparation of the report

Consultation with stakeholders

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Swedish Research Council and National Library Sweden

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

<https://www.vr.se/>, <https://www.kb.se/>

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey



Public



Private



Civil Society



Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN
SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

☒ Yes

☐ No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

Government offices Sweden For institutions and organizations see 1.3

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☒ Yes

☐ No

19. 1.2 cont. **If yes**, which language(s)?

Swedish

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

Here are some awareness raising activities on open science in Sweden 2021-2024: • 2021: The Association of Swedish Higher Education Institutions (SUHF) highlight Recommendation in a Roadmap for Open Science, published in 2021: <https://suhf.se/app/uploads/2024/02/REK-2021-1-Fardplan-for-oppnen-vetenskap-SUHF-Antagen-av-SUHFs-presidium-REVIDERAD-240201-1.pdf> and have also included information in a several relevant webinars. • March 2022: Swedish Research Council (SRC) highlight Recommendation in the report accounting for the work to fulfil the mandate from the Government to promote and coordinate the work of implementing open access to research data, in the description of the contemporary environment: <https://www.vr.se/analys/rapporter/vara-rapporter/2022-03-10-vetenskapsradets-samordningsuppdrag-om-oppnen-tillgang-till-forskningsdata-2022.html> (in Swedish) • March 2022: National Library of Sweden (KB) highlight Recommendation in the report accounting for the work to fulfil the mandate from the Government to promote and coordinate the work of implementing open access to scholarly publications and recommends a national policy on open science: "Samordning av arbete för öppen tillgång till vetenskapliga publikationer. Redovisning av uppdrag 2022": <https://urn.kb.se/resolve?urn=urn:nbn:se:kb:publ-660> (in Swedish) • March 2022: Meeting organised by the Swedish National Commission for UNESCO and Public & Science Sweden (an independent Swedish non-profit membership organisation working to promote dialogue and openness between researchers and the public) with aim to inform about and discuss Recommendation on Open Science with key actors already working actively on open science in Sweden. • April 2022: Presented at webinar reporting on assignments to Swedish Research Council (SRC) and National Library of Sweden (KB) to coordinate activities on open access to research data and scholarly publications, respectively. Recording of webinar: <https://kbplay.mediaflowportal.com/watch/1106070/> (in Swedish) • May 2023: Conference Open Science – from Policy to Practice, conference highlighting different perspectives of shaping, implementing and embedding open science in and beyond the European Union; UNESCO representative participating. Organised by National Library of Sweden, Swedish Research Council, and Public & Science Sweden. Conference web site: <https://oppenvetenskap.se/> • May 2024: Conference presenting the National Guidelines for Open Science. UNESCO representative participating in program. Organised by National Library of Sweden and Swedish Research Council. Recording of conference: <https://kbplay.mediaflowportal.com/watch/1400852/>.

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

General actions taken with regard to open access to scholarly publications: • Funders require immediate open access to publications of results from the research they fund (implemented from 2010 and onwards). • Funders contribute to the national system for funding open access publications via direct partial funding of publication at fully open publishers. • A majority of the funders do not provide individual researchers with grants to APC:s. • Several funders are involved in the discussions on financing a new Open Research Europe (ORE) platform, and have offered support for the work to investigate the possibilities of starting a Nordic Diamond OA competence centre. • Funders have been active partners in the work around the open access system of the future and the Swedish guidelines for open science. General actions taken with regard to open access to research data: In the yearly follow-up of work in Sweden with open access to research data (2024), conducted by the Swedish Research Council, there are funders who have indicated that they do have requirements or incentives for open access to research data (according to the principle "as open as possible, as restricted as necessary") within some calls for grant applications. (Source: Öppen tillgång till forskningsdata – en kartläggning, analys och bedömning 2024 - Vetenskapsrådet) Specific example: As a funder, the Swedish Research Council have the conditions for receiving a grant- to produce a data management plan, (since 2019) and to comply with the Swedish Research Council's guidelines for publication with open access. The requirement regarding open access applies to all types of scientific publications, for example articles, monographs, books and book chapters, for researchers receiving grants as from October 2022. (For more information, see – Publishing open access - Swedish Research Council and Producing a data management plan - Swedish Research Council) A new contextualizing part has been introduced in the application process as a complement to the part that deal with the applicant's competence. In the applicant's resume, the applicant can state "Publications and other research outputs". That means that in the contextualizing part open access to publications and/or research data can also be described, as long as it is relevant for the call in question.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.



Yes



No

25. 2.1 cont. **If yes**

- please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

• A research and innovation bill is presented by the government every fourth year indicating the way forward for Swedish research policy.
<https://www.regeringen.se/contentassets/a70996e2f9de4222b5a6880d33c20d35/forskning-och-innovation-for-framtid-nyfikenhet-och-nytta-prop.-20242560.pdf> In the bill presented in 2024 attention was given specifically to open access to research data and scholarly publications as well as coara. Other documents of relevance: For research infrastructures specifically, there is the Swedish Research Council's guide to research infrastructure 2023, The Swedish Research Council's guide to research infrastructure 2023 - Swedish Research Council, which describes the strategic assessments and recommendations the Swedish Research Council considers to be necessary to ensure that Swedish researchers in all scientific fields can continue to have access to modern, high-quality research infrastructure. One chapter (1.11) in the guide is about open access to research data. • Lund Declaration on Maximising the Benefits of Research Data – <https://www.government.se/information-material/2023/06/lund-declaration-on-maximising-the-benefits-of-research-data/> • Swedish Position Paper on the next EU Research and Innovation Framework Programme, FP10 – <https://www.government.se/information-material/2024/08/swedish-position-paper-on-the-next-eu-research-and-innovation-framework-programme-fp10/>

26. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☒ open science infrastructure
- ☒ open engagement of societal actors
- ☐ open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☒ Yes
- ☐ Partly
- ☐ No

28. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

• National guidelines for Open Science (2024) – <https://urn.kb.se/resolve?urn=urn:nbn:se:kb:publ-738> o Developed by the National Library of Sweden with participation from the Swedish Research Council. o Involved a reference group with researchers as well as representatives from The Agency for Digital Government, The Association of Swedish Higher Education Institutions, Public & Science Sweden, Riksbankens Jubileumsfond, The Royal Swedish Academy of Letters, The Royal Swedish Academy of Sciences, The Swedish Higher Education Authority, The Swedish Library Association, The Swedish National Data Service, The Swedish National Union of Students, The Swedish Research Council, The Young Academy of Sweden. o The guidelines covers the following areas for open science: ☐ Open access to scholarly publications ☐ Open access to research data ☐ Open research methods ☐ Open educational resources ☐ Public engagement in science ☐ Infrastructures supporting open science ☐ There are also sections on actors and their areas of responsibility, as well as on the need for support and guidance. o The guidelines were accompanied by a report giving a national and international background and context for goals and priorities, as well as describing the process for developing the guidelines – <https://urn.kb.se/resolve?urn=urn:nbn:se:kb:publ-721> (In Swedish)

29. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).



Yes



Under development



No

30. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

• Assignment to National Library of Sweden to coordinate, monitor and promote collaboration in the work for open access to scholarly publications and every year publish a report with a combined mapping, analysis and assessment of the national work with open access to scholarly publications. • Assignment to the Swedish Research Council to coordinate, monitor and promote collaboration in the work for open access to research data and every year publish a report with a combined mapping, analysis and assessment of the national work with open access to research data. • Indicators for open access to research data from Swedish Research Council : <https://www.vr.se/english/mandates/open-science/open-access-to-research-data/support-and-tools-/indicators-for-measuring-and-following-up.html> • The Bibsam Consortium – coordinated by the National Library of Sweden and consisting of some 95 higher education institutions and research organisations in Sweden – is used as a lever to achieve a transformation from subscriptions to open access publishing. This is currently achieved through transformative agreements between higher education institutions with multiple publishers as well as agreements with open access publishers at the national level.

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)



Yes



Under development



No

32. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

The National Guidelines for Open Science were adopted and published in January 2024. There are now processes on other levels in the Swedish university system to take the next steps in implementation. The Association of Swedish Higher Education Institutions (SUHF) has developed a Roadmap for Open Science, as well as a guidance for implementing the roadmap: - Roadmap for Open Science (in Swedish) – <https://suhf.se/app/uploads/2024/02/REK-2021-1-Fardplan-for-oppen-vetenskap-SUHF-Antagen-av-SUHFs-presidium-REVIDERAD-240201-1.pdf> - Guidance for implementing the roadmap (in Swedish) – <https://suhf.se/app/uploads/2024/02/Vagledning-for-implementering-av-fardplan-for-oppen-vetenskap-Antagen-av-SUHFs-presidium-REVIDERAD-240201-1.pdf>

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (*ref.: (ii) 17.d, 17.e, 17.g*)



Yes



Under development



No

34. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors). To view the elements of open science, see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9>

As a part of the work with National guidelines on Open Science, the National Library of Sweden has made an inventory of policies for open science and its components at higher education institutions in Sweden (Source: <https://urn.kb.se/resolve?urn=urn:nbn:se:kb:publ-721>). In the yearly follow-up of work in Sweden with open access to research data (2024), conducted by the Swedish research Council, several organisations have indicated that they do have policies for open access to research data (Source: Öppen tillgång till forskningsdata – en kartläggning, analys och bedömning 2024 - Vetenskapsrådet and also in EOSC Monitoring, see answer in 2.9).

35. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ☒ Yes
- ☐ Under development
- ☐ No

36. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

Swedish public research funding organisations (Swedish Research Council, Forte, Formas and Vinnova) in 2023 decided on a funding model to support publication with publishers that exclusively publish fully open access journals. The publishing costs are jointly covered during 2024 and 2025. See - <https://www.kb.se/samverkan-och-utveckling/nytt-fran-kb/nyheter-samverkan-och-utveckling/2023-10-25-funders-in-joint-open-access-initiative.html>

37. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

38. 2.7 cont. **If yes**, please provide details and links as relevant.

Yes, see answer to question 1.4

39. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☐ Yes

☒ No

40. 2.8 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Not aware of any specific efforts being made, and difficulties or obstacles are thus hard to identify.

41. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ☒ Yes
- ☐ Under development
- ☐ No

42. 2.9 cont. **If yes or under development**, please provide details and links as relevant.

Yes – partly. There is no monitoring framework for open science as a whole. However, since 2017 there is a national monitoring on the development of open access to scholarly publications in terms of “Open access in numbers” maintained by the National Library of Sweden. Such indicators are also included in the presentation of the annual budgets from the government. The National Library of Sweden is also assigned to coordinate, monitor and foster cooperation in the area of open access to scholarly publications. The assignment is reported yearly to the government and made openly available. The Swedish Research Council is reporting since 2022 on yearly basis the development of the work on open access to research data, according to indicators for open access to research data (see more information and links in answer on 2.3). Report from 2024: Öppen tillgång till forskningsdata – en kartläggning, analys och bedömning 2024 - Vetenskapsrådet The Swedish Research Council is answering on the annual survey from European Open Science Cloud Steering Board about “the National Contributions to EOSC”, which is connected to the indicators to monitor the EOSC readiness of countries in Europe. Results are presented in the EOSC Observatory, a tool for monitoring policies, practices, and impacts related to the European Open Science Cloud (EOSC): EOSC Observatory The Association of Swedish Higher Education Institutions conduct an annual survey about the work on open science, connected to their Roadmap on open science. See – <https://suhf.se/app/uploads/2024/05/Inventering-2024-Fardplan-for-oppen-vetenskap-Sammanställning-maj-2024.pdf>

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

43. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

3.57% in 2023 (Source: Forskning och utveckling i Sverige 2023 – Preliminär statistik, Statistics Sweden (2024)).

44. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

98.6% of households and companies have access to 1 Gbit/s, or fiber in October 2023. (Source: PTS (Swedish Post and Telecom Authority) "PTS mobiltäcknings- och bredbandskartläggning 2023", 2024-04-04, PTS-ER-2024:13).

45. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

46. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

SUNET (Swedish University Computer Network) – <https://sunet.se/en> SUNET's unique data network and secure and stable IT services enable higher education institutions and other organisations linked to research or higher education to collaborate nationally and globally. SUNET has been a cornerstone of Swedish academia since 1984.

47. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- ☒ Yes
- ☐ Under development
- ☐ No

48. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☒ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☐ community managed infrastructures, protocols and standards, for example those that support biodiversity and engagement with society
- ☐ platforms for exchanges and co-creation of knowledge between scientists and society
- ☒ community-based monitoring and information systems to complement national, regional and global data and information systems

49. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

The Swedish Research Council fund a range of research infrastructures in Sweden. Some of these infrastructures are of relevance with regard to open access and FAIR research data. The research infrastructures can be found here – <https://www.vr.se/english/mandates/research-infrastructure/find-research-infrastructure.html> Open science infrastructures in Sweden of special note : Swepub – National entry for access to Swedish scholarly publications - <https://www.kb.se/samverkan-och-utveckling/swepub.html> Publicera – <https://publicera.kb.se/> National platform for Swedish scholarly open access journals. Developed and maintained by the National Library of Sweden. DiVA – Digitala Vetenskapliga Arkivet – <https://www.diva-portal.org/> DiVA portal is a finding tool and an institutional repository for research publications and student theses used by 50 higher education and research institutions in Sweden. Swedish National Data Service (SND) – <https://snd.se/en> Swedish National Data Service (SND) has a primary function to support the accessibility, preservation, and reuse of research data and related materials. Together with a network of around 40 universities and public research institutes, we form a national infrastructure for research data. SND supports the realization of the national goal of open access to research data by 2026. Funded by the Swedish Research Council. Medborgarforskning.se - <https://medborgarforskning.se/> A Swedish national hub for everyone interested in citizen science. SLU Swedish Species Information Centre, "Artdatabanken" – <https://www.slu.se/artdatabanken/> SciLifeLab – <https://www.scilifelab.se> SciLifeLab is a national resource of unique technologies and expertise available to life scientists, closely intertwined with our community of researchers in areas such as biomedicine, ecology and evolution.

50. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☒ Yes

☐ No

51. 3.5. cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Examples are among the research infrastructures and links mentioned in answer to Q 3.4, for instance: - ICOS Carbon data portal – <https://www.icos-cp.eu/observations/carbon-portal> - European Spallation Source – <https://europeanspallationsource.se/> - National Academic Infrastructure for Supercomputing in Sweden – <https://www.naiss.se/> - Swedish node of Global Biodiversity Information Facility – <https://www.gbif.se/gbif-sweden/> - Swedish NREN, Sunet, <https://www.sUNET.se/> (mentioned in answer to Q3.3) and NORDUnet – <https://nordu.net/> Other examples of relevance: - Worldwide LHC Computing Grid – <https://home.cern/science/computing/grid> - International Bathymetric Chart of the Arctic Ocean – <https://ibcao.org> - Data from the EISCAT Scientific Association – <https://eiscat.se/scientist/data/> - Data from the Swedish Institute of Space Physics – <https://www.irf.se/sv/om-irf/data/> - The Human Protein Atlas – <https://www.proteinatlas.org> - SIPRI databases – <https://www.sipri.org/databases> - The SKA Observatory, Regional Centres – <https://www.skao.int/en/science-users/119/ska-regional-centres>

52. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☐ Yes

☒ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

53. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☒ Yes

☐ No

54. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

The purpose of the Roadmap for Open Science, developed and adopted by the Association of Swedish Higher Education Institutions (SUHF) is to clarify the responsibilities and actions needed by higher education institutions in Sweden in order to transition to an open science system. The goal is to contribute to creating better possibilities for the organisations and other actors to coordinate around common issues and to encourage collaboration. The recommended actions are all intended to build capacity in higher education and/or research performing institutions in Sweden in a systematic manner. The roadmap and actions taken are continuously monitored, and the roadmap is revised occasionally. This capacity building initiative is formalized through decisions by the general assembly of SUHF. See Roadmap for Open Science (in Swedish) – <https://suhf.se/app/uploads/2024/02/REK-2021-1-Fardplan-for-oppen-vetenskap-SUHF-Antagen-av-SUHFs-presidium-REVIDERAD-240201-1.pdf> Both formal and informal capacity building activities are performed at individual organisations, sometimes in cooperation between them.

55. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

56. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

57. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☐ Yes

☒ No

58. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

• 18 higher education institutions in Sweden are working with the Human Resources Strategy for Researchers (HRS4R) framework, which includes open science. HRS4R is the European Commission's initiative for the development of researchers' working conditions and career development frameworks. • PhD courses on open science is developed at several higher education institutions, see for example: - Stockholm university – <https://www.su.se/doctoral-school-in-the-humanities/courses/open-science-for-phd-students-5-ects-1.651401> - Linköping university – <https://biblioteket.blog.liu.se/2024/10/21/new-course-open-science-for-phd-students/> - Umeå university – <https://www.umu.se/en/humlab/education/postgraduate-studies/data-management-and-open-science/>

59. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☐ Yes
- ☒ Partly
- ☐ No

60. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ☒ Yes
- ☐ No

61. 4.5 cont. **If yes**, please provide more information and links as relevant.

The Swedish Research Council is mandated by the government to coordinate science communication, and work with a long-term approach to ensure research benefits society, and encourage dialogue and collaboration between researchers and other societal actors. For further information, see – <https://www.vr.se/english/mandates/science-communication.html> The website forskning.se publishes news about research from Swedish universities, research institutes, and research organisations. The Swedish Research Council coordinates the development of forskning.se, and the editorial team is based at the agency. The website also offers a service that helps journalists get in touch with researchers and a guide that explains how research works (Fatta forskning). Visit the web site – <https://www.forskning.se/> Public & Science Sweden (Vetenskap & Allmänhet), an independent Swedish non-profit membership organisation works to promote dialogue and openness between researchers and the public. Visit Public & Science Sweden – <https://vetenskapallmanhet.se/eng/>

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

62. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☒ Yes

☐ No

63. 5.1 cont. **If yes**, please provide more information and links as relevant.

The Association of Swedish Higher Education Institutions (SUHF) has a working group on merits and evaluation: <https://suhf.se/arbetsgrupper/arbetsgruppen-for-meritbedomningar/> Specific example: in the Swedish Research Councils' position on assessment of researcher merits research data is mentioned: <https://www.vr.se/soka-finansiering/bedomning-av-ansokningar/sa-gar-bedomningen-till.html#:~:text=St%C3%A4llningstagande%20g%C3%A4llande%20bed%C3%B6mningen%20av%20forskares%20meriter>

64. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☒ Yes

☐ No

65. 5.2 cont. **If yes**, please provide details and links as relevant.

In a report the Swedish Research Council together with a team of researchers and communication experts presented a new framework for courses in science communication at doctoral level. These courses aim to enhance researchers' abilities to communicate their findings and engage with the public and other stakeholders outside academia. Several universities have already implemented the course in science communication. See report – <https://www.vr.se/english/analysis/reports/our-reports/2022-09-29-framework-for-courses-in-science-communication.html>

66. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☒ Yes

☐ No

67. 5.3 cont. **If yes**, for each initiative, please provide more information, including on the stakeholders[i] that are involved or lead the initiative, and links as relevant.

[i] For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

Several institutions in Sweden have signed the Agreement on Reforming Research Assessment from the Coalition for Advancing Research Assessment (CoARA) and are planning to work in accordance with it. See information about the Swedish chapter of the CoARA initiative – <https://coara.eu/coalition/national-chapters/coara-national-chapter-sweden/> A number of awards for science communication have been established, with the aim of rewarding and recognising researchers who make significant efforts to engage with stakeholders outside academia and/or the public. For example prizes are awarded by Örebro University and the Hamrin foundation, Young Academy of Sweden, Forum for Science Communication (organised by several Swedish funding bodies) and the SNS, Center for Business and Policy Studies.

68. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☐ Yes

☒ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

69. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

- ☒ YES, at the national level
- ☐ YES, at the institutional level
- ☐ YES, at both the national and institutional levels
- ☐ No

70. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☐ promotion of pre-prints
- ☐ exploration of open peer-review practices
- ☐ valuing and sharing of negative scientific results and associated data
- ☒ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☐ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☐ promoting open innovation practices
- ☐ Other

71. 6.1. cont. **If yes**, please provide more information and links as relevant

The Association of Swedish Higher Education Institutions (SUHF) has addressed some aspects of it regarding research process generally in recommendations of Roadmap for Open Science. See – <https://suhf.se/app/uploads/2024/02/REK-2021-1-Fardplan-for-oppen-vetenskap-SUHF-Antagen-av-SUHFs-presidium-REVIDERAD-240201-1.pdf> The National Library of Sweden in 2023 completed a survey and analysis of public participation in the scientific process, with a focus on libraries, as part of a mission from the government. The report includes recommendations to actors to promote public participation in science. See the report – <https://urn.kb.se/resolve?urn=urn:nbn:se:kb:publ-720> The website Medborgarforskning.se collects citizen science projects in Sweden and is connected to the EU Citizen Science initiative. See – <https://medborgarforskning.se/>

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

72. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Primarily through participation in different networks dedicated to open science. On a European level Sweden is active in the European Open Science Cloud (EOSC), OpenAIRE, Open Research Europe (ORE), Council for National Open Science Coordination (CoNOSC), International Coalition of Library Consortia (ICOLC), Open Access 2020, among others. The National Library of Sweden and/or higher education institutions also supports a number of infrastructures for open science/access, including: arXiv.org, Crossref, COAR, Counter, DOAJ, DOAB, OASPA, OA Switchboard, Open Journal Systems (PKP), ORCID, OurResearch, SciPost, SPARC Europe, etc.

73. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☒ Yes

☐ No

74. 7.2 cont. **If yes**, please provide links as relevant and more information.

The Swedish Research Council is mandated by the Swedish government to coordinate and promote Swedish engagement in the European Open Science Cloud (EOSC), which is connected to international scientific collaborations (among other things through best practices, capacity building and metrics for open science). Other national strategies and plans for stimulating cross-border multi-stakeholder collaboration on open science among other things: - Swedish strategy for participation in Horizont Europa 2021-2027 – <https://www.regeringen.se/informationmaterial/2021/10/en-nationell-strategi-for-svenskt-deltagande-i-horisont-europa-20212027/> - Lund declaration on maximising the benefits research data (June, 2023) – <https://www.government.se/information-material/2023/06/lund-declaration-on-maximising-the-benefits-of-research-data/> - Swedish Position Paper on the next EU Research and Innovation Framework Programme, FP10 – <https://www.government.se/information-material/2024/08/swedish-position-paper-on-the-next-eu-research-and-innovation-framework-programme-fp10/>

75. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☒ Yes

☐ No

76. 7.3 cont. **If yes**, please provide links as relevant and more information.

Several Swedish public research funding organisations are involved in the discussions on funding mechanisms to finance the Open Research Europe (ORE) platform in Europe.

Part 8. GENERAL CONSIDERATIONS

77. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☐ Yes

☒ No

78. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org